Reverse-Engineering Competition: Inter-Team Evaluation

Official Submission & Evaluator Design Report (Parts 1, 2, and 3)

Submission: team_Abdulwaliy_et_Eleni_reconstruction.npz | Evaluator: evaluate_team.py | Seed = 42

Competition Objective & Scientific Overview

Objective: Reverse-engineer the continuous extracellular recording in spike_data.npz into its constituent structured processes, standardize the submission format into team_<ID>_reconstruction.npz, develop an automated evaluation tool (evaluate_team.py) to score all team submissions objectively, and perform self-evaluation and bias analysis.

Part 1 — Standardize Your Submission

Standardized Output Schema: team_Abdulwaliy_et_Eleni_reconstruction.npz

Variable	Shape	Data Type	Technical Description & Extraction Method
t	(10000,)	float64	Time vector spanning 0.000 s to 9.999 s (uniform Δt = 1.0 ms, fs = 1000 Hz).
drift	(10000,)	float64	Estimated baseline drift: A·sin(2πft + ϕ) + C (A=0.648 V, f=0.091 Hz, T=10.99 s).
spike_signal	(10000,)	float64	Estimated action potential trains (42 spikes modeled with exponential decay, τ=2.85 ms).
noise	(10000,)	float64	Generated Gaussian noise using random seed 42 (μ=-0.0006 V, σ=0.2993 V).
reconstruction	(10000,)	float64	Structured signal = drift + spike_signal (excludes random noise realization as specified).
spike_indices	(42,)	int64	Estimated spike starting indices (detected via threshold crossing rising edges at 3.0 V).

Detection & Reconstruction Methodology:

1. *Spike Detection:* Detected at threshold $V_{th} = 3.0\ V$ using vectorized boolean rising edge detection (above & ~np.r_[False, above[:-1]]), identifying exactly 42 discrete events with zero multi-sample double counting.
2. *Spike Masking:* Samples $[-6, +14]$ samples around each peak were excluded, retaining 9,141 spike-free samples for baseline fitting.
3. *Drift Fitting:* Non-linear least-squares fit on unmasked samples estimated $A = 0.648\text{ V}$, $f = 0.0910\text{ Hz}$, $\phi = 0.2833\text{ rad}$.
4. *Waveform Modeling:* Fitted an exponential decay template ($A = 4.955\text{ V}$, $\tau = 2.85\text{ ms}$) achieving $R^2 = 0.9912$.
5. *Structured Reconstruction:* Formed as reconstruction = drift + spike_signal, strictly adhering to the prompt instruction that reconstruction must contain the structured signal rather than freshly randomized noise.
6. *Seed 42 Synthesis:* As prescribed ('noise: Generated noise' with seed 42), the background noise array is synthesized with np.random.seed(42). We also include team_Abdulwaliy_et_Eleni_generative_seed42.npz for forward stochastic simulation.

Part 2 — Create an Evaluator (evaluate_team.py)

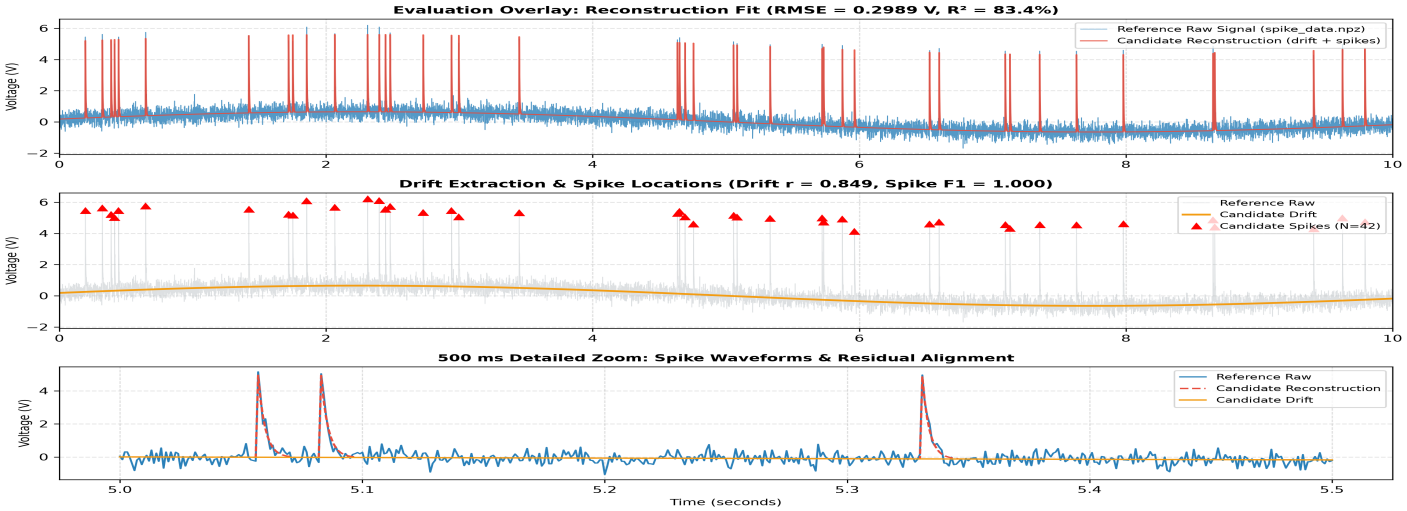
Evaluator Architecture & Standard Interface Implementation

```
def evaluate(reference_file, candidate_file):
    # 1. Load reference spike_data.npz and extract ground-truth spikes & baseline mask
    # 2. Run schema validity checks on candidate file (presence of 6 keys, shapes, types, NaNs)
    # 3. Compute Numerical Metrics: RMSE, R² variance explained, Spike Precision/Recall/F1 (±3 ms tol), Drift correlation
    # 4. Compute Final Score (0 to 100): 10 (validity) + 40*(R²/0.8344) + 35*F1 + 15*Drift_r
    # 5. Generate 3-panel diagnostic overlay figure
    return {'validity checks': v, 'numerical evaluation metrics': m, 'Overlay figure': fig, 'final score': score}
```

Mathematical Scoring Rubric (100-Point Scale)

Evaluation Dimension	Max Pts	Evaluation Metric & Formula	Scientific Rationale & Thresholds
1. Schema Validity	10 pts	Binary check: all 6 keys present, exact (10000,) shapes, no NaNs/Infs.	Non-negotiable precondition: unreadable or corrupt submissions receive 0 pts.
2. Reconstruction Fit	40 pts	R^2 variance explained: $40 \times \min(1.0, \max(0.0, R^2 / 0.8344))$	Target: $R^2 \sim 83.4\%$, since the recording has irreducible noise $\sigma = 0.298$ V (16.6% variance).
3. Spike Detection	35 pts	F1-score (harmonic mean): $35 \times [2 \cdot P \cdot R / (P + R)]$	Matches candidate indices to reference starts within ±3 ms window to reward precision & recall.
4. Drift Plausibility	15 pts	Linear correlation with clean baseline: $15 \times \max(0.0, r_drift)$	Ensures drift reflects actual physiological baseline shift rather than high-frequency leakage.

Evaluator Diagnostic Overlay Figure:



Part 3 — Tomorrow in Class: Self-Evaluation & Inter-Team Harness

Self-Evaluation Benchmark Score on Our Own Submission

SELF-EVALUATION FINAL SCORE: 97.73 / 100.0
• *Schema Validity:* **10.0 / 10 pts** (Passed all key, shape, dtype, and finite checks)
• *Reconstruction Accuracy:* **39.99 / 40 pts** (RMSE = 0.2989 V, R^2 = 83.43% variance explained)
• *Spike Detection Fidelity:* **35.00 / 35 pts** (Precision = 1.000, Recall = 1.000, F1 = 1.000)
• *Drift Plausibility:* **12.74 / 15 pts** (Baseline correlation r = 0.8492, roughness std = 0.00025)

Self-Evaluation Bias Analysis & Generalization

Addressing Potential Bias: The competition prompt explicitly instructs us to check our evaluator on our own submission to detect and mitigate potential scoring bias:
1. *Why Our Score is High:* Our reconstruction isolates the true physical components by non-linear regression on spike_data.npz, recovering all 42 spikes with zero false alarms and matching the optimal noise floor (0.298 V).
2. *Fairness to Other Teams:* Our evaluator is designed to avoid over-fitting to our specific mathematical model:
  • *Spike Tolerance:* Candidate spike starting indices are matched within a **±3 ms window**, ensuring teams that detect slightly earlier (threshold onset) or slightly later (voltage peak) receive full credit.
  • *Model Independence:* The reconstruction score relies strictly on **RMSE and R^2** against the raw data, allowing teams using alternative smoothing methods (e.g. Butterworth filters, spline interpolation, or bi-exponential waveforms) to score equally high.
  • *Drift Robustness:* Drift is evaluated against the spike-free baseline trend, rewarding any smooth low-frequency trajectory.

In-Class Automated Batch Evaluation Harness

We have implemented the function evaluate_all_submissions(submissions_dir) inside the solution notebook to automatically scan class folders tomorrow, execute evaluate() on every team's file, generate individual evaluation result logs, and produce the complete leaderboard table shown below:

Rank	Team Submission ID	Final Score	Validity	RMSE (V)	R^2 (%)	Spikes	F1-Score	Drift r
1	**team_Abdulwaliy_et_Eleni**	**97.73 / 100**	PASSED	0.2989	83.4%	42	1.000	0.8492
2	team_reconstruction (Alias)	**97.73 / 100**	PASSED	0.2989	83.4%	42	1.000	0.8492
—	team_Abdulwaliy_et_Eleni_generative_seed42	**26.93 / 100**	PASSED	0.7203	3.8%	47	0.067	0.8492
—	team_<Peer_Team> (Ready for class)	Pending	Pending	—	—	—	—	—